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CSCI 543/DATA 532 - Machine Learning - Project Report

Title: Predicting future values for monthly electricity consumption

Acknowledgement: While working on our project, we encountered some difficulty with obtaining the radar dataset. Our sincere thanks to our Professor, **Dr. Meera Gopinath Sujatha** for her guidance to evaluate the performance of different regression models on electricity consumption data. This project would not have been possible without the support of our teammate Ms. Taylor Dolan, who built the FFNN model and helped us interpret the model results with her domain expertise. Sid built the SVM and LSTM models. Vig was responsible for overall project management, code testing and providing feedback to Taylor/Sid, doing the literature review, drafting the report and making slides.

Abstract

This class project aims to apply Machine Learning (ML) regression models to predict monthly electricity consumption for two climatically distinct U.S. cities, Minneapolis and Phoenix. The dataset used is a subset of the monthly electricity consumption dataset for metropolitan cities in the USA by Dr. Obringer from Penn State [1]. Monthly socioeconomic data from the United States Census Bureau is also taken as inputs for the models. Feature engineering was used to capture weather-energy relationships. The result is a feature-rich dataset incorporating temperature, dew point, solar radiation, seasonal encodings, and lagged values. Three supervised machine learning models, Feed Forward Neural network (FFNN), Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) were trained to predict monthly electricity consumption for both cities. Model evaluation was done using standard regression metrics like the correlation coefficient (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and data visualization techniques. Time Series Split cross-validation is applied to assess temporal generalization and model stability across different time periods. To ensure interpretability, Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) are used to identify the most influential features and to explain individual predictions. Results show that LSTM achieves the best overall performance in Phoenix, while SVM has the best performance for Minneapolis. The model performance comparison along with interpretability analysis provides accurate forecasts and actionable insights for energy planning. This project is an early exploration effort to be used as a reference for further research.

Section A: Introduction

Inspiration: This project idea came from Vig and Taylor's PhD dissertations, which involve applying AI/ML techniques to space radar and meteorological data. We have used real electricity consumption data to build and test three ML models which Taylor needed to move her dissertation forward.

Goals: This endeavor seeks to explore how three supervised machine learning models perform when predicting monthly electricity consumption. We originally considered doing both classification and regression on multiple datasets, but we narrowed our scope to a single regression problem to better highlight the strengths and weaknesses of each model. The dataset includes monthly electricity usage and meteorological variables for two major U.S. cities, Minneapolis and Phoenix. We compare three regression models: FFNN, SVM and LSTM.

Expected Outcomes:
Our effort aims to compare the performance of several machine learning models on an actual dataset rather than filling in a specific research gap. Based on our theoretical understanding from the class, our null hypothesis is that the LSTM will not be effective in predicting monthly electricity consumption.

H0: SVM or FFNN will be the most effective models for forecasting electricity consumption

H1: LSTM will outperform SVM and FFNN for forecasting electricity consumption

To compare the performance of these models, we evaluate each model using common regression metrics such as correlation coefficient (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), as well as k-fold cross-validation on the training set.

Project Significance:
With cities facing extreme weather and growth in population, accurately forecasting electricity demand is important for utilities, city planners and energy researchers. Understanding how temperature and seasonal patterns affect electricity consumption is critical. One possible approach to this big data problem is machine learning (ML). These models execute quickly, identify patterns and learn data relationships. This project offers the data science community and stakeholders several significant contributions:

- ☐ **Model Comparison:** We evaluate three different supervised ML models to understand which approach is most effective for time-series energy forecasting.
- ☐ **Feature Engineering:** We demonstrate how weather variables, seasonal encodings, and lagged features can improve electricity-use prediction.
- ☐ **Explainability:** We use SHAP and LIME to interpret model behavior and highlight which features most strongly influence predictions.
- ☐ **Cross-Validation Use:** We apply Time Series Split to measure how stable each model is when forecasting across different time periods. This provides confidence that the models generalize well after the training window.

Section B: Literature Review

Machine learning has been successfully applied in meteorology and energy forecasting to enhance predictive performance on nonlinear data. Support Vector Machines (SVMs) have proven effective for temperature and precipitation forecasting, demonstrating strong capability in modeling complex atmospheric relationships [2], [3]. Hasan et al. [4] showed that Support Vector Regression outperformed traditional forecasting methods for rainfall prediction using six years of historical data, while Wang et al. [5] improved temperature prediction accuracy by optimizing SVM regression, raising the coefficient of determination from ~0.8 to ~0.9. Collectively, these studies highlight the value of kernel-based SVM approaches for nonlinear meteorological datasets.

El-Azab et al. [6] examined hourly electricity demand from the ISO New England region throughout 2021, incorporating seasonal segmentation, temperature, calendar effects, and market variables. Using Adaptive Neuro-Fuzzy Inference System (ANFIS), LSTM, Gated Recurrent Unit (GRU), and Artificial Neural Network (ANN) models, they found that LSTM and ANFIS consistently achieved the highest R^2 values across all seasons. Similarly, Abumohsen et al. [7] analyzed hourly electrical load data from the Tubas Electricity Company in Palestine and compared LSTM, GRU, and Recurrent Neural Network (RNN) models using an 80/20 split. GRU achieved superior performance on R^2, MSE, and MAE, demonstrating enhanced capability for capturing temporal dependencies—supporting the importance of evaluating models with these metrics.

Dudek [8] investigated load forecasting across multiple countries with datasets exhibiting triple seasonality (annual, weekly, and daily cycles), integrating calendar effects and temperature. Random Forest was benchmarked against Autoregressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), Prophet, Multilayer Perceptron (MLP), SVM, ANFIS, LSTM and XGBoost. The study concluded that Random Forest outperformed both classical statistical models and advanced machine learning techniques in this context.

Cui et al. [9] utilized the Residential Energy Consumption Survey (RECS) dataset to predict Energy Use Intensity (EUI) using Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost) and XGBoost. SHAP analysis provided interpretability, identifying common influential household characteristics and supporting the development of targeted, practical energy-efficiency strategies for the U.S. residential sector.

Across the literature, SVMs [2][3][4][5], LSTMs [6][7][8] and neural network–based models [6][7][8] are consistently used for forecasting tasks involving nonlinear or temporal data. Interpretability methods such as SHAP [9] further enhance model transparency and provide valuable insight into feature importance. These findings motivated us to use SVM, LSTM and FFNN models along with explainability tools such as SHAP, LIME and validation techniques like k-fold cross-validation.

Section C: Methodology

Dataset details: The Obringer dataset was used and monthly electricity usage was collected from January 2007 through December 2018, with the timeframes slightly varying between datasets [1]. For this project, we focused on two U.S. cities: Minneapolis, MN and Phoenix, AZ. The dataset contains several years of monthly historical electricity consumption expressed in megawatt-hours (MWh/capita), making it suitable for regression and time-series forecasting tasks. Each city's dataset was handled separately but processed using identical steps so performance comparisons across models would be fair.

Dataset attributes: The Minneapolis data frame had 96 records (rows) with 53 features (columns), while the Phoenix data frame had 114 records (rows) with 47 features (columns), as the humidity index was not available. They contain monthly timestamps along with numerical attributes from the socioeconomic and meteorological data. Since the goal of the project is forecasting future electricity consumption, the dependent feature is the monthly electricity consumption, used to create supervised learning labels. All features were numerical; therefore, no one-hot or label encoding was needed.

Methodology: For this electricity forecasting project, several key steps are involved:

- *Visualizing the Data:* The first step was to import the Minneapolis and Phoenix datasets as Pandas data frames and visualize their structure using shape, size, head, tail, info(), describe() and time-series line plots. These visualizations showed city-specific seasonal trends in electricity usage and guided the model selection process.
- *Preprocessing:* Data cleaning included checking for missing or duplicated entries. Any missing monthly consumption values were imputed using forward fill to maintain continuity. After basic cleaning, the model was run just with the meteorological and socioeconomic variables. After the first run, feature

engineering was performed to enrich the dataset. Feature engineering variables included lag variables at one, six, and twelve months. This feature is a past value of the variable and allows for the capture of temporal dependencies. These engineered features helped improve the models' ability to capture external factors influencing electricity demand.

- *Re-visualizing the Data*: After preprocessing and feature engineering, additional plots such as correlation heatmaps and distribution plots were generated. These visualizations confirmed that scaling was successful and showed meaningful relationships across electricity usage, meteorological variables, and socioeconomic features.

Machine Learning Models for regression:

Three supervised regression models were used to forecast monthly electricity consumption: Support Vector Machines (SVM), Feedforward Neural Network (FFNN), and Long Short-Term Memory Network (LSTM). SVM with its mathematical calculation of the support vectors in the hyperplane is good for non-linear regression, FFNN with its many hidden layers and deep learning capability identifies relationships between features and LSTM is good for seasonality trends.

- *Splitting the dataset:* For each city, the dataset was divided into training and testing subsets using a standard 80/20 split. For LSTM, the split was done before windowing to avoid introducing future-data leakage.
- *Training & Testing:* Training and test data were kept separate to prevent overfitting. The SVM and FFNN models were trained using .fit() and evaluated on the test set. The LSTM model required reshaping the input into a three-dimensional tensor (samples × timesteps × features) before training. Predictions were generated based on the test set to assess model generalization.
- *Regression metrics:* Standard regression metrics such as MAE, RMSE and R^2 were computed for each model to quantify forecasting accuracy. Learning curves were also plotted for neural network models to evaluate convergence and training behavior.

Section D: Results and Discussion

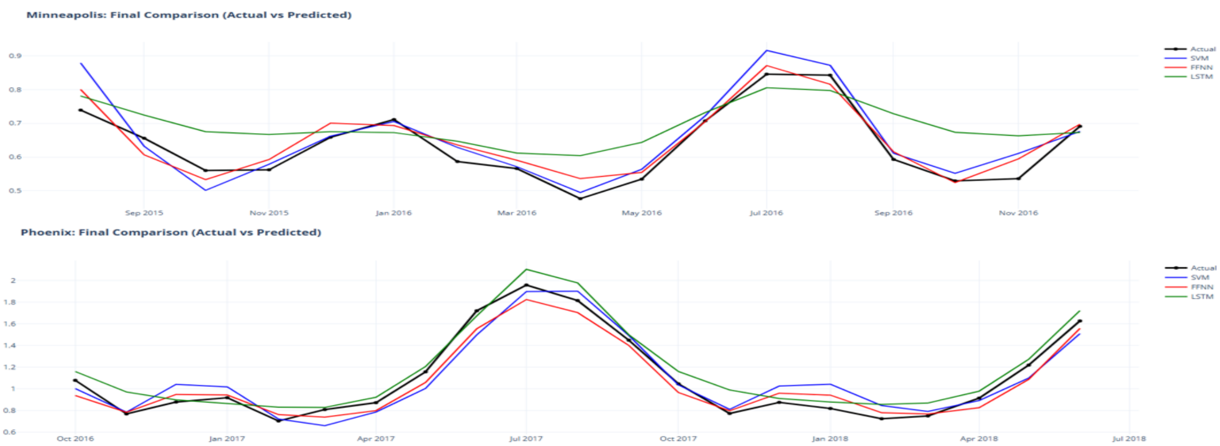
Initial Challenges and Approach: We faced several challenges related to forecasting electricity consumption for Minneapolis and Phoenix. Unlike a typical classification task, this was a regression and time-series prediction problem where patterns vary by climate zone. Minneapolis exhibits extreme seasonal variation, while Phoenix shows high year-round temperatures with different usage patterns. Although the same preprocessing and feature engineering methods were used for both cities, these contrasting patterns made the forecasting task more complex. Selecting the right models was another challenge. To ensure a fair comparison, we implemented three distinct models—SVM, FFNN, LSTM and evaluated them on almost identical, standardized datasets for each city.

Support Vector Machines (SVM): SVM *Regression (SVR)* attempts to fit a function within an ε-insensitive margin, minimizing prediction error while keeping the model complexity low. After scaling the input features, SVR was first trained on the training dataset and evaluated using the test set. Cross validation was performed afterward to assess how stable the model's performance was across different splits of training data.

Feed Forward Neural Network (FFNN): The FFNN model was selected for its deep learning capability. After defining the network architecture with dense hidden layers and ReLU activation, the model was trained using forward as well as back propagation and evaluated using the same k-fold structure. We used k=5 as that gives us a better idea of model performance. FFNN captured more complex feature interactions but performance varied depending on the city.

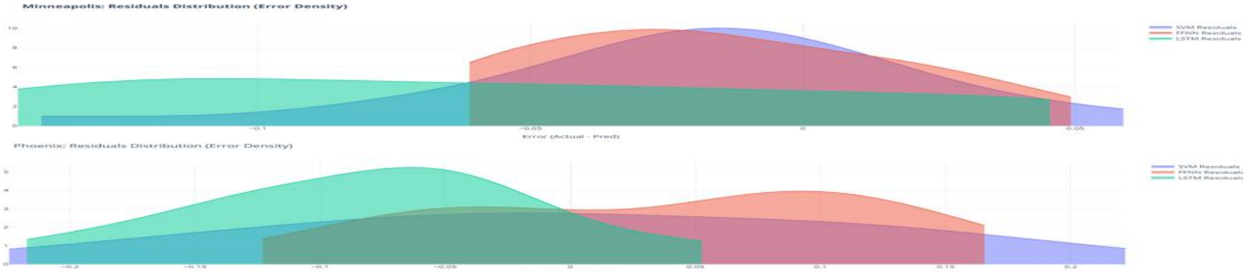
Long Short-Term Memory (LSTM): The LSTM model was fed 12-month windows of historical data. Reshaping the dataset into three-dimensional tensors (samples × timesteps × features) enabled the LSTM to retain memory across many months of observations. The model learned longer-term trends, seasonality patterns and lag-dependent relationships more effectively than SVM and FFNN.

Figure 1 shows how the 3 ML models predicted each month's electricity consumption and how it compared to the actual values for Minneapolis and Phoenix.



.Figure 1: Actual Vs. Predicted value for the three ML models

Model Evaluation and Comparison: After training all three models, we compared their performance using standard regression metrics: MAE, RMSE, and R^2. Visualizations from the Google Colab notebook including the correlation heatmap, loss curves, and side-by-side bar charts comparing each model's MAE/RMSE/R^2 allowed for a clear evaluation on how well each model predicted electricity consumption for each city. For the Minneapolis dataset, the best performing model was SVM and for Phoenix, the best performing model was LSTM. The second-best performing models were the FFNN for Minneapolis and SVM for Phoenix. Figure 2 shows how the residuals were distributed for the three ML models across the two cities. There is a slight left skew for Minneapolis (long tail on the right) and a more normal like distribution for Phoenix.



.Figure 2: Residual distribution for the three ML models

Explainable AI (xAI) Analysis: To interpret each model's predictions, we applied SHAP and LIME to the trained models. SHAP provided global feature importance, confirming that lagged variables were dominant drivers of predictions. LIME supplied local explanations for individual predictions, illustrating how specific conditions influenced the output. These xAI tools validated that the engineered features were meaningful and improved interpretability across all three models. Figure 3 shows the SHAP results and Figure 4 shows the LIME results for both cities. SHAP results indicate that lagged electricity consumption and temperature related features are the most influential predictors for both cities. LIME explanations demonstrate how individual predictions are driven by a small number of key features, providing local interpretability of the FFNN model. For Minneapolis, LIME showed that lagged features such as last year's electricity load and the six month previous load tend to reduce the electricity prediction, indicating the model is recognizing strong seasonality. For Phoenix, last year's electricity load and temperature are pushing the prediction upward, indicating Phoenix's energy demand and temperature driven nature.

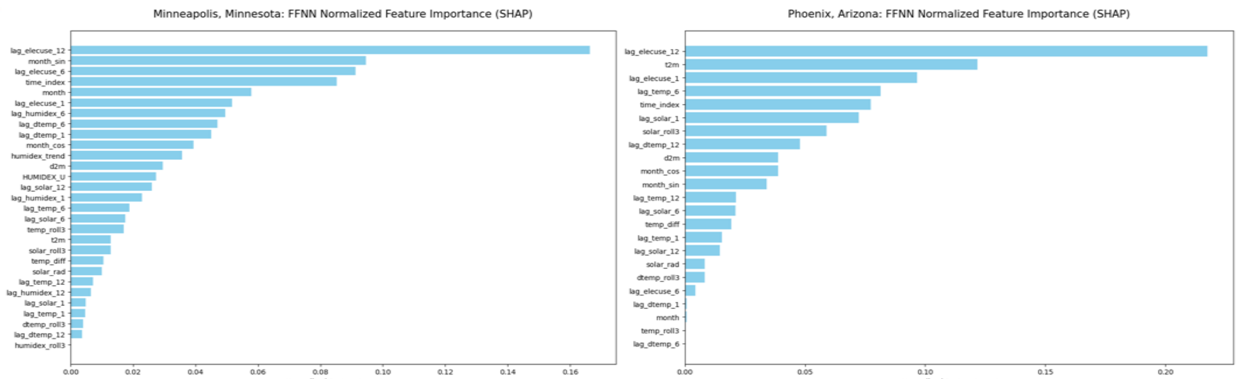


Figure 3: SHAP results for Minneapolis and Phoenix

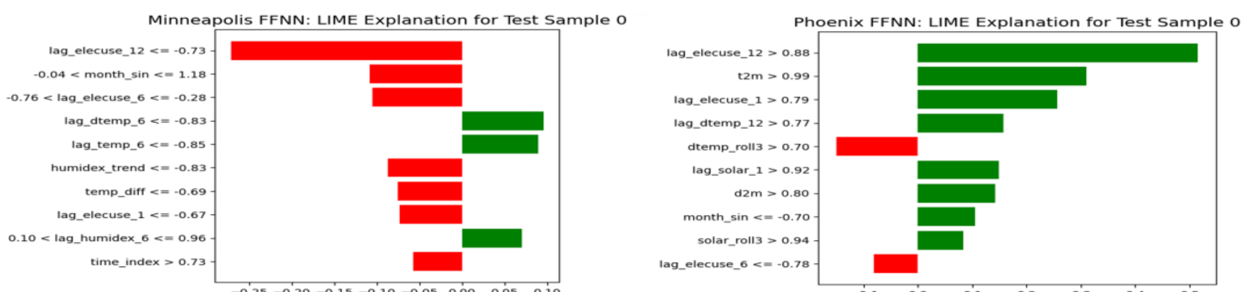


Figure 4: LIME results for Minneapolis and Phoenix

Section E: Hypothesis Definition and Testing

To determine which model performed best for electricity consumption forecasting, we defined two hypotheses:

- H_0 (Null Hypothesis): SVM or FFNN will be the best performing model.
- H_1 (Alternative Hypothesis): LSTM performs better.

Evaluation Metrics:
To evaluate how the models performed, we rely on established regression metrics such as MAE, RMSE, R^2 and k-fold cross-validation averages. These metrics ensured statistically grounded model comparisons without the need for custom scoring formulas.

Hypothesis Test Findings:

In Table 1 and Table 2, evaluation metrics are shown for Phoenix and Minneapolis for each model from the holdout data. From Arizona (Table 1), the lowest errors across all models are seen from LSTM. From Minnesota (Table 2), the lowest errors across all models are seen from SVM.

Phoenix, Arizona	MAE	RMSE	R^2
SVM	0.09	0.11	0.9
FFNN	0.1	0.13	0.87
LSTM	0.05	0.07	0.96

Table 1: Evaluation Metrics for Phoenix, Arizona for all models

Minneapolis, Minnesota	MAE	RMSE	R^2
SVM	0.03	0.04	0.79
FFNN	0.04	0.04	0.77
LSTM	0.04	0.05	0.69

Table 2: Evaluation Metrics for Minneapolis, Minnesota for all models

When comparing to the five-fold cross validation for Minneapolis as seen in Figure 5, SVM is still the best performing model. From Figure 6 for Phoenix, the best performing model becomes SVM due to the lowest errors.

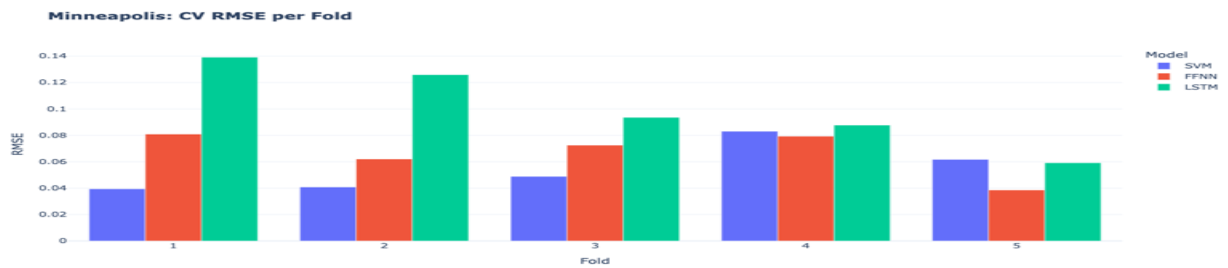


Figure 5: Five-fold cross validation for Minneapolis, MN

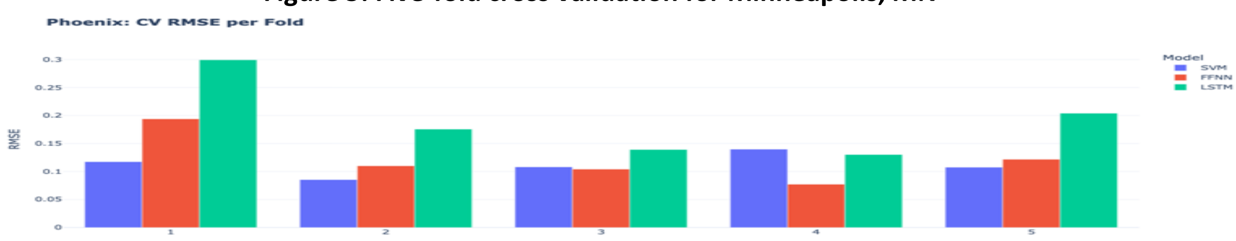


Figure 6: Five-fold cross validation of RMSE for Phoenix, AZ

Because our time series prediction prioritizes performance on unseen data, we consider the holdout set (the remaining 20% of unseen data) to be more representative of real-world forecasting. Given these findings, we had to accept each hypothesis based on the given city. For Minneapolis, we rejected H1 and accepted H0 as SVM outperformed the FFNN and LSTM. SVM is known to work best with less noisy datasets, indicating the city's less extreme seasonal shifts, capturing the overall trend without overfitting. For Phoenix, we rejected H0 and accepted H1 as LSTM outperformed the other models. Since Phoenix has a more extreme seasonal variation, LSTM can better capture temporal dependencies and patterns.

Section F: Conclusion

Good things:
This project provided valuable hands-on experience applying machine learning and deep learning techniques to a real-world energy forecasting problem. We became comfortable working in Python and implementing multiple modeling approaches, including SVM, FFNN and LSTM. The project reinforced important concepts such as regression modeling, time-series learning, cross-validation, and model evaluation using standard performance metrics. We also gained practical insight into the importance of feature engineering and the use of xAI tools (SHAP, LIME) to improve model interpretability.

Challenges faced:
An early challenge was the difference in electricity consumption patterns between Minneapolis and Phoenix, which required careful consideration of seasonality and climate effects. Preparing the data for time-series modeling, especially reshaping inputs into tensors for LSTM, was technically challenging. Selecting the best models and metrics to evaluate their performance requires a deeper understanding of the model's inner workings. Lastly, verifying that the models were learning meaningful patterns instead of overfitting highlighted the importance of domain knowledge.

Future Scope:
We were constrained by time for this class project but there is more work that can be done in this research area. Some of our ideas include testing the models on other cities and changing the time series period from monthly to weekly or daily if possible. Literature shows researchers have tried hybrid architectures (eg. CNN–LSTM models) which could also be explored in the future.

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